

NHAI HACKATHON 7.0

EDGEAUTH AI

Offline Facial Recognition & Liveness
Authentication System

By Shivangi Kakkar



PROBLEM STATEMENT

- No Connectivity
 - Remote highway zones lack internet access
- Cloud Dependency
 - Existing auth systems require online access
- Spoofing Risk
 - Photo-based attendance fraud is easy
- Edge AI Gap
 - Existing solutions are not optimized for fully offline edge environments.

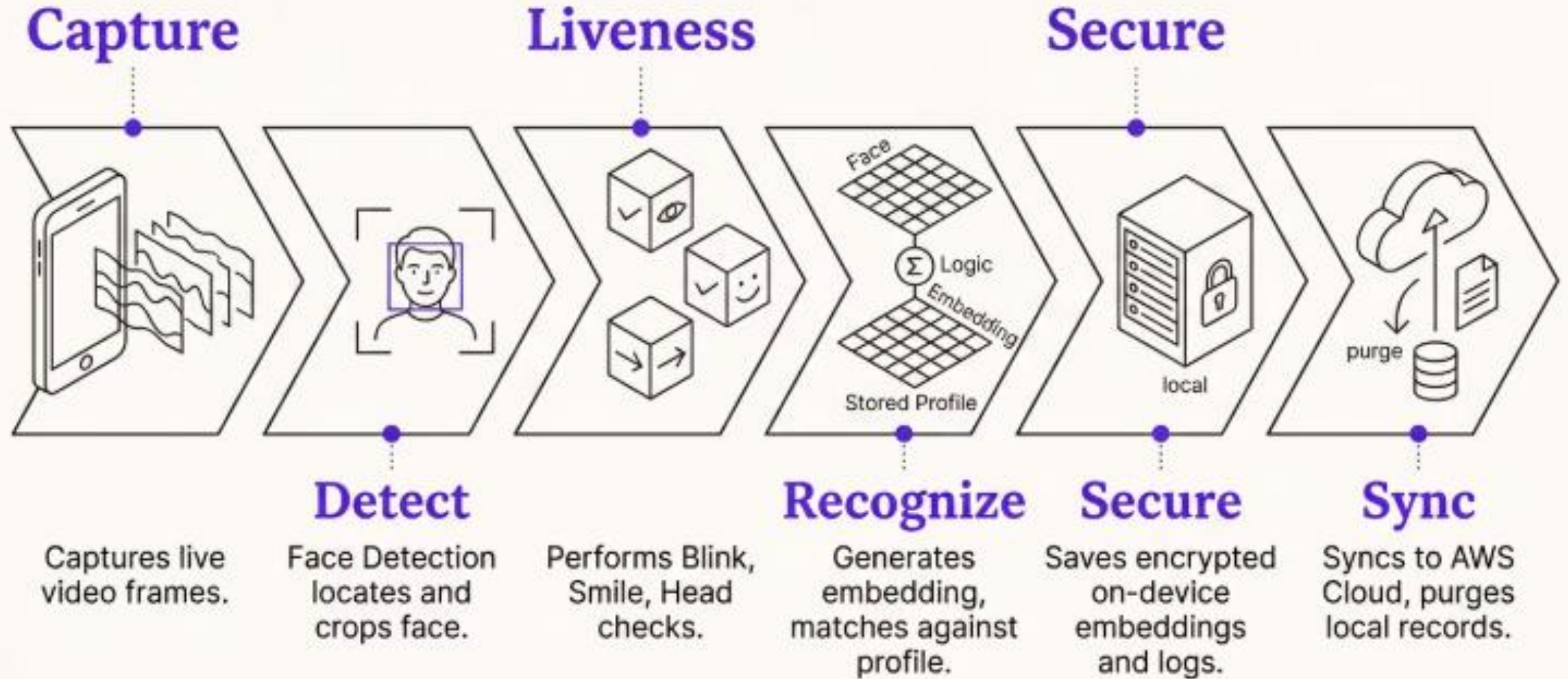
OUR SOLUTION

Designed specifically for low-connectivity highway and field operations.

- Offline-First Auth
- Liveness Check
- Local Secure Storage
- AWS Sync

A fully offline, privacy-first authentication pipeline built for zero-network field operations

ARCHITECTURE DIAGRAM



KEY FEATURES



Offline Face Recognition

Identifies users without internet



Head Movement Check

Liveness via pose



AWS Sync

Auto-sync when online



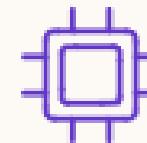
Blink Detection

Anti-spoofing via eye



Secure Local Storage

Encrypted SQLite on device



TFLite Models

Optimized for low-end devices

TECHNOLOGY STACK

Component

- Frontend
- Face Recognition
- Liveness Detection
- ML Runtime
- Local Storage
- Cloud Sync

Technology

- React Native
- MobileFaceNet
- MediaPipe
- TensorFlow Lite
- SQLite
- AWS

INNOVATION & ADVANTAGES

<1s

Inference Time

Real-time on-device recognition

<20MB

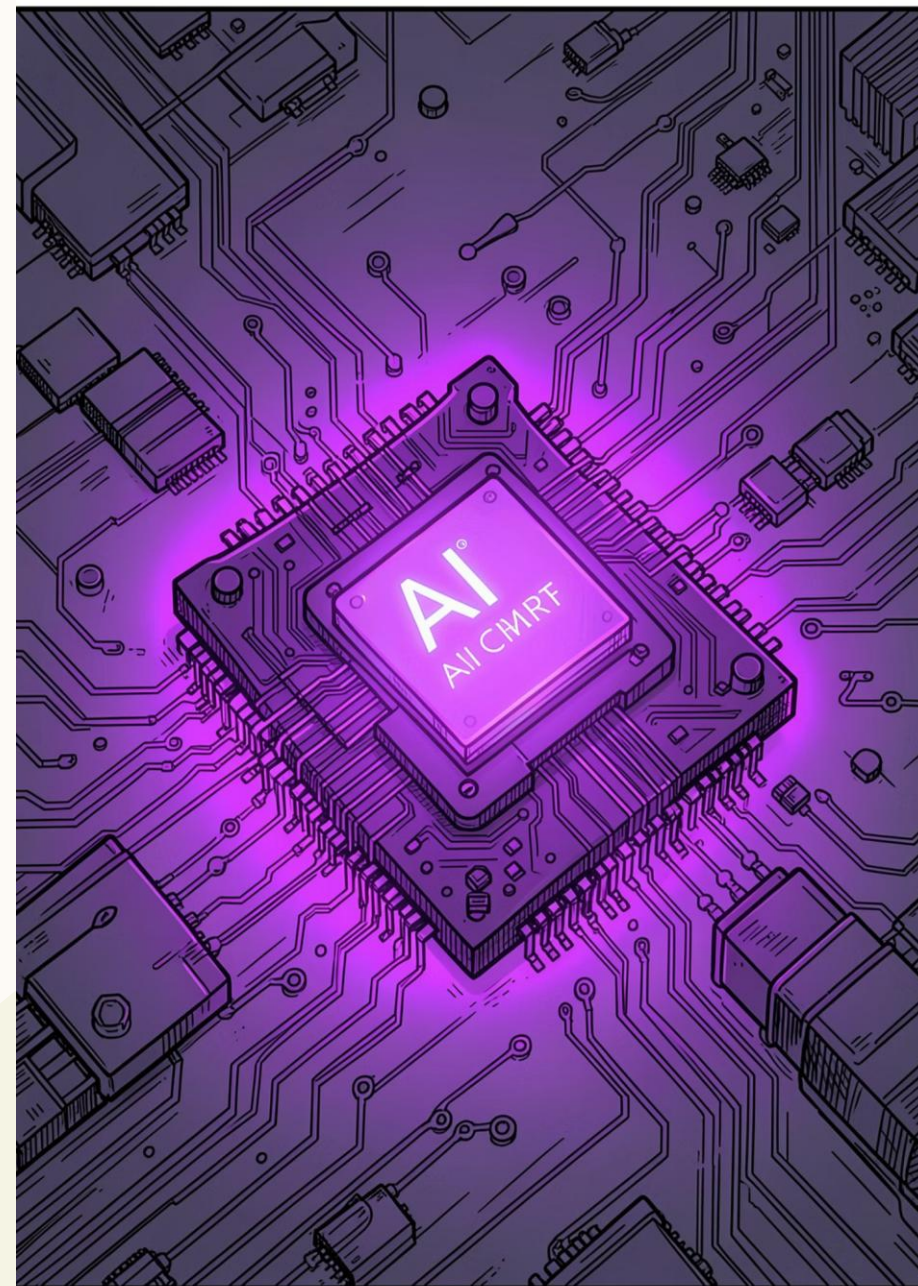
App Footprint

Lightweight architecture target

0%

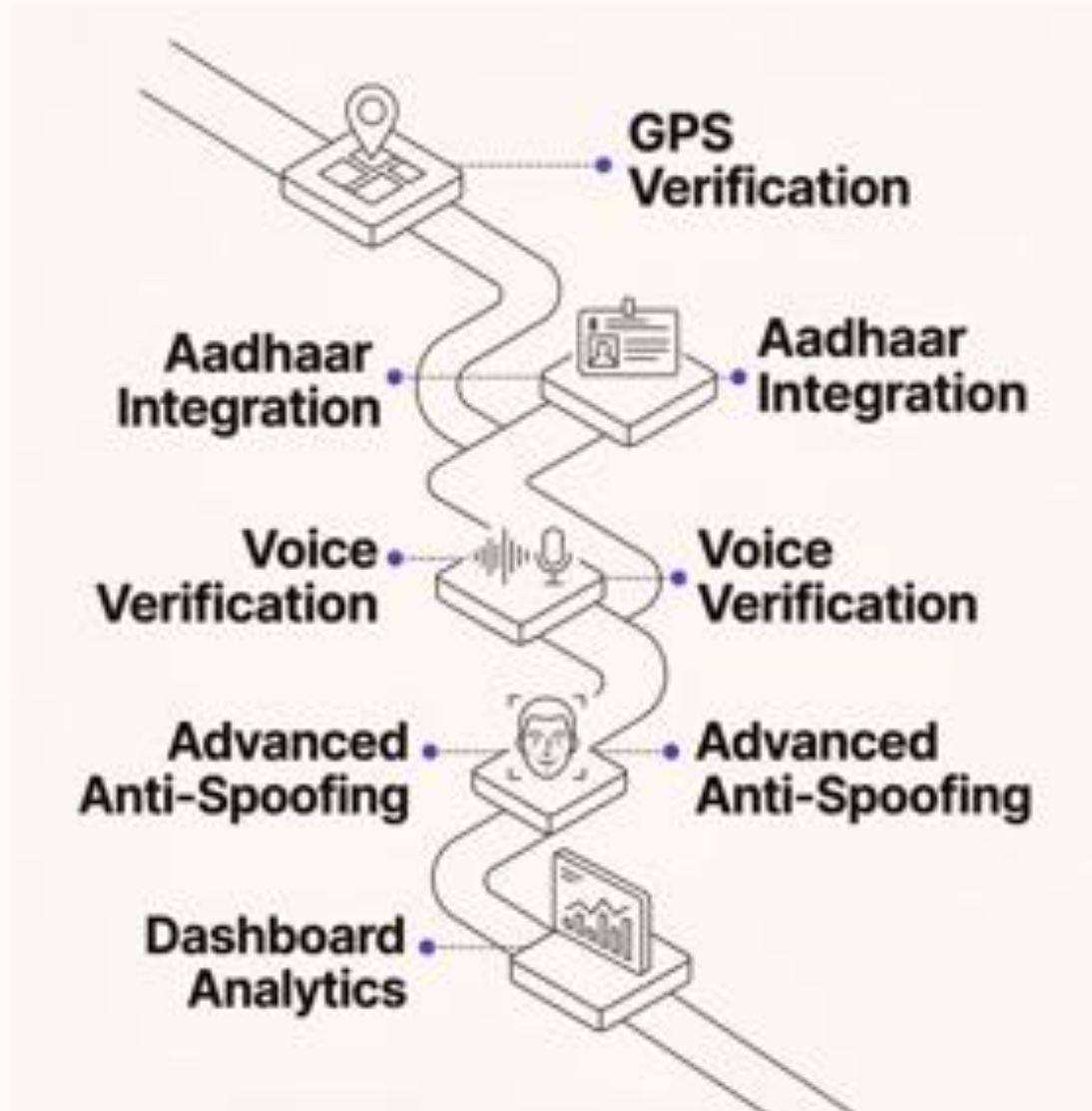
Internet Required

Fully offline-first design



FUTURE SCOPE

8

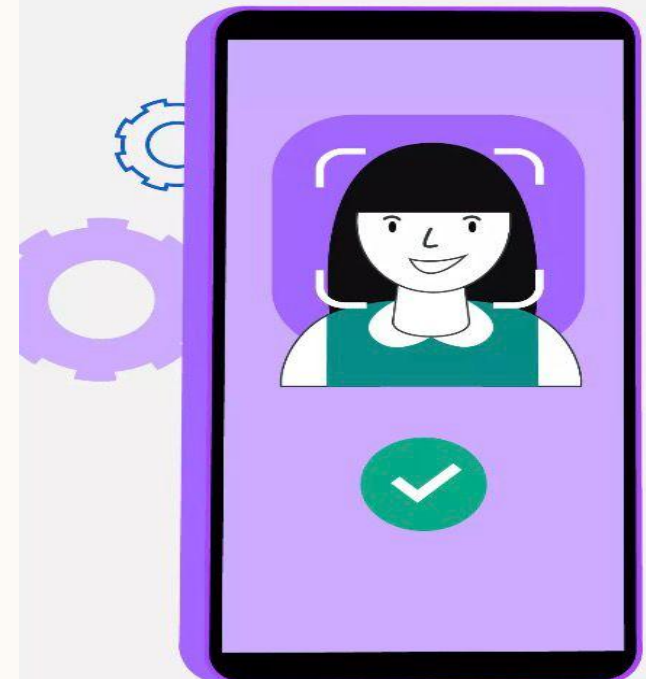


Next Milestones

EdgeAuth AI evolves into a multi-modal, integrated field authentication platform.

CONCLUSION

EdgeAuth AI delivers a **lightweight, secure, and scalable** authentication solution for remote field operations — **zero network required**.



THANK YOU

EdgeAuth AI – NHAI Hackathon 7.0

Shivangi Kakkar

- Offline-First
- AI-Powered
- Field Ready